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Accurate and lightweight oral cancer detection using SE-MobileViT on clinically validated image dataset

Md Firoz Kabir[1], Md Yousuf Ahmad[2], Roise Uddin[3], Martin Cordero[1] and Shashi Kant[4*]

*Correspondence:
Shashi Kant
drshashikant38@gmail.com
[1]University of the Cumberlands, Williamsburg, USA
[2]Trine University, Angola, USA
[3]Pacific States University, Los Angeles, USA
[4]Bule Hora University, Hagere Maryam, Ethiopia

Abstract

Oral cancer poses a critical global health challenge, with early detection significantly improving patient survival rates and treatment outcomes. This study proposes an advanced deep learning-based diagnostic model, LightSE-MobileViT, specifically designed to classify oral cancer using medical imaging. The Oral Cancer Classification dataset used in this study comprises clinically validated lip and tongue images collected from various ENT hospitals in Ahmedabad. The original dataset consisted of 131 images (87 cancerous and 44 non-cancerous). To address class imbalance and enhance model generalizability, data augmentation techniques were employed, expanding the dataset to 981 images with equal distribution across both classes. Our proposed model, LightSE-MobileViT, integrates a lightweight convolutional neural network (CNN) backbone consisting of sequential convolutional layers enhanced with batch normalization and rectified linear unit activations. To further enrich feature representation and spatial attention, a Squeeze-and-Excitation block is embedded after the third convolutional layer. Subsequently, a MobileViT transformer encoder is employed, effectively capturing global contextual information through efficient multi-headed self-attention mechanisms. Experimental evaluations revealed that LightSE-MobileViT achieved superior diagnostic performance, attaining an accuracy of 98.39%, precision and recall values approaching 1.00 for both cancerous and non-cancerous categories, a macro F1-score of 0.98, and an ROC-AUC of 1.00. Comparative analysis demonstrated notable improvements over benchmark models, including CST-CNN (98% accuracy), MobileNetV2 (97% accuracy), DenseNet121 (97% accuracy), and InceptionV3 (90% accuracy). The exceptional performance of LightSE-MobileViT underscores its robust capability and clinical applicability, suggesting significant potential for deployment in automated oral cancer screening, thus facilitating early detection and timely intervention.

Keywords Oral cancer detection, Lightweight deep learning, MobileViT transformer, Squeeze-and-excitation (SE) module, Medical imaging

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1 Introduction

Oral cancer represents a major global health burden, ranking among the most prevalent and fatal forms of cancer worldwide. According to the World Health Organization (WHO), approximately 377,713 new cases and 177,757 deaths are reported annually, underscoring its substantial morbidity and mortality rates. Early and accurate detection remains critical to improving survival outcomes, as late-stage diagnosis significantly diminishes treatment efficacy and prognosis.

Conventional diagnostic methods for oral cancer primarily include visual inspection, biopsy, histopathological analysis, and imaging modalities such as CT and MRI. Although widely used, these approaches are often invasive, expensive, time-consuming, and prone to observer variability, potentially delaying timely diagnosis and intervention. This highlights the urgent need for reliable, non-invasive, and efficient diagnostic tools to facilitate early detection and improve patient management. Advances in artificial intelligence, particularly in DL, have revolutionized computer-aided diagnosis (CAD) systems, offering powerful tools for medical imaging analysis. Among these, CNN-based models have demonstrated outstanding performance in classification and segmentation tasks by automatically learning hierarchical representations from medical images. However, standard CNN architectures often struggle to capture complex global relationships within images, limiting their diagnostic precision.To overcome these challenges, hybrid CNN-Transformer models have emerged, combining the strong local feature extraction capabilities of CNNs with the global contextual understanding of Transformer networks. Motivated by these advancements, this study proposes a novel hybrid model, LightSE-MobileViT, specifically designed for efficient and precise classification of oral cancer images. The architecture integrates lightweight CNN layers, SE blocks for channel-wise feature recalibration, and MobileViT modules to capture long-range dependencies alongside local textures, thereby enhancing feature representation.The primary objective of this research is to develop and evaluate the LightSE-MobileViT model for distinguishing between cancerous and non-cancerous images of the oral cavity. To ensure robust and balanced training, we utilized an augmented version of the publicly available Oral Cancer Classification (OCI) dataset, originally consisting of 131 clinically validated images (87 cancerous and 44 non-cancerous), collected from ENT hospitals in Ahmedabad. Through data augmentation, the dataset was expanded to 981 images, with equal representation of both categories. This enhanced dataset enabled a comprehensive evaluation of the proposed model's effectiveness in addressing the diagnostic challenges of oral cancer.

The contributions of this paper include:

- Introducing the LightSE-MobileViT model, a lightweight yet powerful hybrid CNN-Transformer architecture.
- Comprehensive validation on augmented clinical images, demonstrating excellent performance and robustness.
- Providing a reliable, efficient, and clinically applicable CAD tool for early oral cancer detection, with substantial potential to enhance diagnostic precision and patient outcomes.

This work underscores the transformative potential of advanced deep learning techniques in medical diagnostics, paving the way for more accurate, accessible, and timely cancer care.

2 Related work

Sampath et al. [1] proposed a deep learning fusion strategy integrating ResNet-50 features with logistic regression to classify oral cancer from lips and tongue images, achieving a high accuracy of 97.8% through hierarchical feature fusion and SGD optimization. Bansal et al. [2] performed a comparative analysis of machine learning classifiers including Neural Networks, KNN Ensemble, and SVM on oral cancer datasets, highlighting the effectiveness of segmentation and feature extraction in enhancing classification accuracy. Mannelli et al. [3] introduced a defect-based classification system for tongue cancer resection, offering a structured surgical approach to optimize functional and oncologic outcomes through stratified defect types and tailored reconstructions.

Warnakulasuriya et al. [4] presented a WHO-endorsed consensus updating the classification of oral potentially malignant disorders (OPMDs), incorporating new entities and emphasizing standardized nomenclature for improved diagnosis and risk assessment. Alabi et al. [5] compared machine learning models to predict locoregional recurrence in early-stage oral tongue cancer, with the Boosted Decision Tree model achieving 81% accuracy, outperforming traditional DOI-based prognosis. Muller and Tilakaratne [6] reviewed WHO's 2022 updates on oral cavity and mobile tongue tumors, detailing restructured sections and new inclusions like non-neoplastic lesions, enhancing the precision of clinical and pathological diagnoses. Shamim et al. [7] demonstrated that transfer learning with deep convolutional neural networks can classify five types of tongue lesions with near-human performance, supporting automated pre-screening of oral cancer in low-resource settings.

Ansarin et al. [8] proposed a five-tier classification system for glossectomy procedures based on surgical anatomy and tumor spread, establishing a unified terminology to improve surgical communication and research consistency. Bansal et al. [9] introduced a CNN-based model named Oral_Cancer_Detection, trained on a small Kaggle dataset, achieving 94% accuracy in classifying lips and tongue images as cancerous or non-cancerous with low complexity. Dwivedi et al. [10] developed a fusion-based deep learning model using augmented lip and tongue images, achieving 94.62% accuracy and outperforming several state-of-the-art models, demonstrating its reliability for early-stage oral cancer detection. Rajaguru and Prabhakar [11] compared GMM, MLP, and ELM classifiers for oral cancer classification using TNM staging data, with GMM showing the highest average accuracy of 94.18%, emphasizing its clinical diagnostic potential.Müller [12] reviewed the 2017 WHO classification highlighting the exclusion of oropharynx, new entries like schwannoma, and the streamlined categorization of oral tumors, aiding clarity in pathological diagnosis. Zini et al. [13] analyzed 36 years of national cancer data, identifying site-specific incidence and survival trends; tongue and gums had the lowest survival rates, reinforcing the need for targeted early screening and site-based risk assessment. Bagan et al. [14] detailed the clinical presentation of OSCC, noting its typical painless onset in high-risk sites like the tongue and floor of the mouth, and the need for differential diagnosis from other aggressive oral malignancies. Speight and Farthing [15]discussed pathological distinctions between oral and oropharyngeal cancers,

highlighting HPV's role and prognostic factors, which are essential for developing targeted therapeutic strategies. Sciubba [16] emphasised the epidemiological distinctions between lip and intra-oral cancers, noticing a decline in the age-adjusted incidence of lip cancer in contrast to a continuous increase in intra-oral cancer rates. The study emphasises the necessity of a distinct classification system to facilitate precise diagnosis and targeted treatment strategies. Sciubba [16] underscored the importance of early diagnosis in enhancing the survival rates of oral cancer, focussing on key etiological factors such as tobacco, alcohol, betel nut use, and HPV. He elucidated the progression from dysplasia to invasive squamous cell carcinoma and emphasised that surgery is the primary treatment. Welikala et al. [17] utilised ResNet-101 and Faster R-CNN to create deep learning models for the automated detection and classification of oral lesions as part of the MeMoSA project. Their system demonstrated the feasibility of low-cost, AI-assisted early oral cancer screening by achieving F1 scores of 87.07% for lesion identification and 78.30% for referral classification.

3 Dataset and Preprocessing

The dataset employed in this study consists of oral cavity images specifically focusing on the lips and tongue, categorized into two classes: cancerous and non-cancerous. These images were originally collected from various ENT hospitals in Ahmedabad, India, and subsequently underwent clinical validation by experienced ENT specialists to ensure diagnostic integrity.The initial dataset comprised 131 images—87 labeled as cancerous and 44 as non-cancerous [18–20]. Due to the inherent class imbalance and the limited size of the original dataset, extensive data augmentation was applied to improve model generalizability and prevent overfitting. Augmentation was performed using the Keras ImageDataGenerator, applying a range of geometric and photometric transformations including random rotations (up to 30°), scaling (± 20%), horizontal and vertical shifts (± 20%), horizontal flipping, and nearest-neighbor pixel filling[21–23].This augmentation process expanded the dataset to a total of 981 images, resulting in approximately 500 images per class and achieving balanced representation. All images were resized to a standardized dimension of 256 × 256 pixels, and pixel values were normalized to a [0, 1] range to ensure stable convergence during convolutional neural network training [24, 25].This comprehensive preprocessing pipeline ensured the availability of a high-quality and balanced dataset, enabling effective training and evaluation of the proposed LightSE-MobileViT model for oral cancer classification.

3.1 The SE-MobileViTNet

In this study, we present a novel lightweight hybrid architecture named SE-MobileViTNet for oral cancer diagnosis. Our model leverages sequential CNN blocks for hierarchical feature extraction, channel-wise attention using Squeeze-and-Excitation (SE) blocks, and global feature dependency modeling via a MobileViT-based Transformer Encoder. This architecture is optimized for the Oral Cancer Classification (OCI) dataset, using only two class labels: cancer and non-cancer.As illustrated in Fig. 1, the input to the model is a high-resolution RGB image of the oral cavity with a dimension of 256 × 256 × 3[26–28]. The image is passed through four convolutional blocks (Conv1–Conv4), each consisting of a 3 × 3 Conv2D layer, followed by BN and a ReLU activation[29, 30].

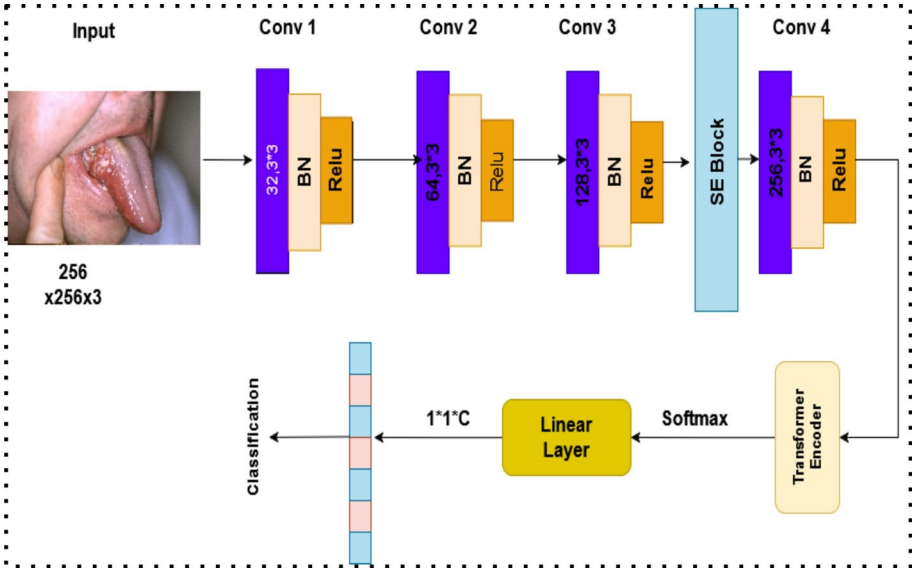


Fig. 1 Architecture of the proposed LightSE-MobileViT model for oral cancer classification

Mathematically, the feature extraction can be defined as:

$$SF_1 = ReLU\left(BN\left(I * W_1\right)\right) \tag{1}$$

$$SF_2 = ReLU\left(BN\left(SF_1 * W_2\right)\right) \tag{2}$$

$$SF_3 = ReLU\left(BN\left(SF_2 * W_3\right)\right) \tag{3}$$

3.2 Squeeze-and-excitation (SE) block

After the third convolution block,we introduce a Squeeze-and-Excitation (SE) block to refine channel-Wise dependencies and recalibrate feature responses, replacing the LGA used in LWENet Fig. 1. The SE block performs:

- Squeeze: Global average pooling for each channel C:

$$Z_c = \frac{1}{H * W} \sum_{I=1}^{H} \sum_{J=1}^{W} SF3\left(c\right)\left(i, j\right) \tag{4}$$

- Excitation: Two FC layers with ReLU and sigmoid activations:

$$\delta = \sigma\left(W_2 * ReLU\left(W_1 * z\right)\right) \tag{5}$$

- Recalibration: Channel- Wise multiplication:

$$S\text{^}F_3^{(c)} = \delta_c * SF_3^{(c)} \tag{6}$$

- This recalibrated feature map is than passed to ConV4:

$$SF_4 = ReLU\left(BN\left(S\text{^}F_S * w_4\right)\right) \tag{7}$$

To address long-spatial dependencies, the feature map SF4 is passed through a MobileViT block,which replaces the high-cost AMSA block in the original LWENet. This encoder is designed with:

LayerNorm -MobileViTblock- Residual Add
LayerNorm -MLP– Residual Add

$Let\ F\ \in R^{H*W*C}$ be the output of Con4. Then the encoding process is:

$$z_1 = MobileViT\left(Norm\left(SF_4\right)\right) + SF_4 \tag{8}$$

$$Z_2 = MLP\left(Norm\left(z_1\right)\right) + z_1 \tag{9}$$

3.3 Classification Head

The Output from the MobileViT block is globally averaged and passed through a fully connected (linear) layer with Softmax activation for final binary classification:

$$F_{avg} = \frac{1}{H * W}\sum_{i=1}^{H}\sum_{j=1}^{W} Z_2\left(i, j\right) \tag{10}$$

$$P_{out} = \frac{e^{F_{avg}}}{\sum_{j=1}^{C} eF_{avg_j}} \tag{11}$$

3.4 Proposed model architecture: LightSE-MobileViT

The proposed LightSE-MobileViT model is specifically engineered to accurately classify oral cancer using advanced deep learning techniques. The architecture integrates sequential convolutional neural network (CNN) layers, Squeeze-and-Excitation (SE) blocks, and a lightweight MobileViT transformer encoder to achieve a robust balance between local and global feature extraction.Initially, the input RGB oral cavity images, resized to $256 \times 256 \times 3256\backslash times\ 256\backslash times\ 3256 \times 256 \times 3$ dimensions, are passed through four convolutional blocks (Conv1–Conv4). The first convolutional layer employs 32 filters of size $3 \times 33\ \backslash times\ 33 \times 3$, capturing fundamental spatial patterns. Subsequent convolutional layers apply 64, 128, and 256 filters respectively, while retaining the same kernel size of $3 \times 33\backslash times\ 33 \times 3$, progressively enhancing hierarchical feature representation. To ensure stable and effective training, each convolutional layer is followed by Batch Normalization (BN) and a Rectified Linear Unit (ReLU) activation [31, 32].After the third convolutional block, a Squeeze-and-Excitation (SE) block is incorporated to recalibrate channel-wise feature responses. The SE block employs global average pooling (squeeze operation) followed by two fully connected layers activated by ReLU and sigmoid functions (excitation operation), producing adaptive channel weights to emphasize informative features and suppress redundant ones [32, 33]. This enhances the model's diagnostic precision by selectively focusing on discriminative patterns within the oral cavity images.Subsequent to the convolutional and SE operations, a MobileViT transformer encoder block is integrated (see Fig. 2) to capture long-range dependencies and global contextual information. The design of the MobileViT block is as follows:

Patch Extraction: Feature maps from the SE block are partitioned into non-overlapping 2×2 patches, which are then flattened into input tokens.

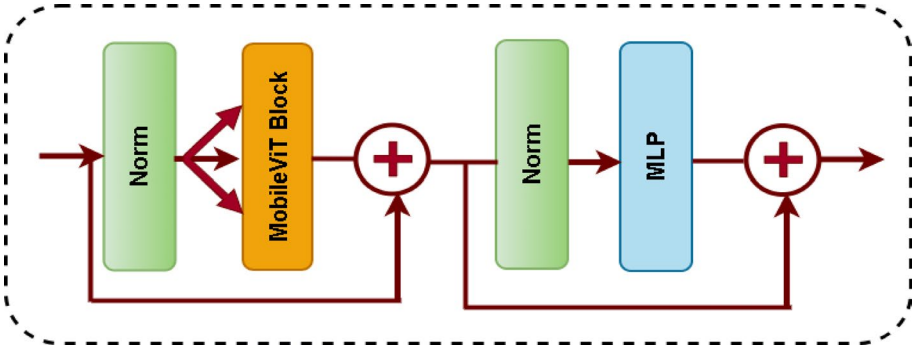


Fig. 2 Internal structure of the MobileViT block used in LightSE-MobileVit

Token Embedding: Each patch is linearly projected into a 128-dimensional embedding space to form the input for the transformer encoder.

Transformer Configuration: A single transformer encoder block with 4 self-attention heads is utilized, striking a balance between model compactness and computational expressiveness [34, 35]. Each head attends to different parts of the token sequence to model complex relationships.

Feed-Forward Network (FFN): The transformer encoder includes an FFN comprising two dense layers:

- First layer with 256 units and GELU activation,
- Second layer with 128 units matching the embedding size.

Positional Encoding:

Learnable positional embeddings are added to the input tokens before the attention mechanism to retain spatial information lost during flattening.

Normalization and Residuals:Both the multi-head attention and the FFN layers are followed by residual connections and Layer Normalization, facilitating stable gradient flow and improving convergence speed [36, 37].After transformer processing, the output tokens are reshaped back to the spatial domain and concatenated with the original convolutional feature maps, thus preserving and integrating both local textures and global contexts for robust classification.The final aggregated feature map is passed through a Global Average Pooling (GAP) layer to reduce spatial dimensions, followed by two dense layers with 128 and 64 neurons, each applying ReLU activation and dropout regularization (dropout rate = 0.5) to prevent overfitting. Finally, a fully connected output layer with softmax activation produces the probability scores for the two classes: cancerous and non-cancerous.This hybrid architecture, combining lightweight convolutional processing and transformer-based contextual learning, offers significant potential for clinically deployable, interpretable, and efficient oral cancer detection (Fig. 2).

3.5 Loss function

To ensure reproducibility and clarity, this section outlines the detailed experimental setup used in developing and evaluating the LightSE-MobileViT model.For the dataset, we design a combined loss function using Binary Cross- Entropy(BCE) and Contrastive Loss(CL).Binary Cross- Entropy Loss:

$$L_{BCE} = -\frac{1}{N}\sum_{i=1}^{N}\left[x_i log\left(y_i\right) + \left(1 - x_i\right) log\left(1 - y_i\right)\right] \tag{12}$$

Contrastive Loss:

$$L_{CL} = \frac{1}{2N}\sum_{i=1}^{N}\sum_{j=1}^{N}\left[x_{ij} d_{ij}^2 + \left(1 - x_{ij}\right) max\left(0, m - d_{ij}\right)^2\right] \tag{13}$$

where:xi is the true label, yi is the predicted probability,dij is the distance between samples i and j,m is the contrastive margin (e.g., 1).We empirically set α=β=0.5.

Final Loss:

$$L_{OCI} = \alpha * L_{BCE} + \beta * L_{CL} \tag{14}$$

3.6 CST-CNN

We introduce a novel convolutional neural network architecture named Multi-Scale Residual Fusion CNN (MSRF-CNN) for accurate classification of oral cancer Fig. 3. The model is tailored to extract discriminative features from tongue and lip images by leveraging multi-scale spatial representation, residual-inspired parallel paths, and deep fusion mechanisms. This model is built upon principles similar to the CST-YOLO backbone but customized for image classification.

Stage 1: Initial Convolutional Block,the first Convolutional block applies 32 filters of size 3*3 followed by ReLU activation and max pooling, producing feature map SF1:

$$SF_1 Maxpool\left(ReLU\left(Conv2D_{32,3*3}\left(I'\right)\right)\right) \ldots . \tag{15}$$

Stage 2: Multi-Scale parallel Convolution,this block consists of three parallel convolutional paths with different kernel sizes(3*3, 5*5, and 7*7), simulating receptive field diversity:

$$P_1 = MaxPool\left(ReLU\left(Conv2D_{64,3*3}\left(SF_1\right)\right)\right) \tag{6}$$

$$P_2 = MaxPool\left(ReLU\left(Conv2D_{64,5*5}\left(SF_1\right)\right)\right) \tag{17}$$

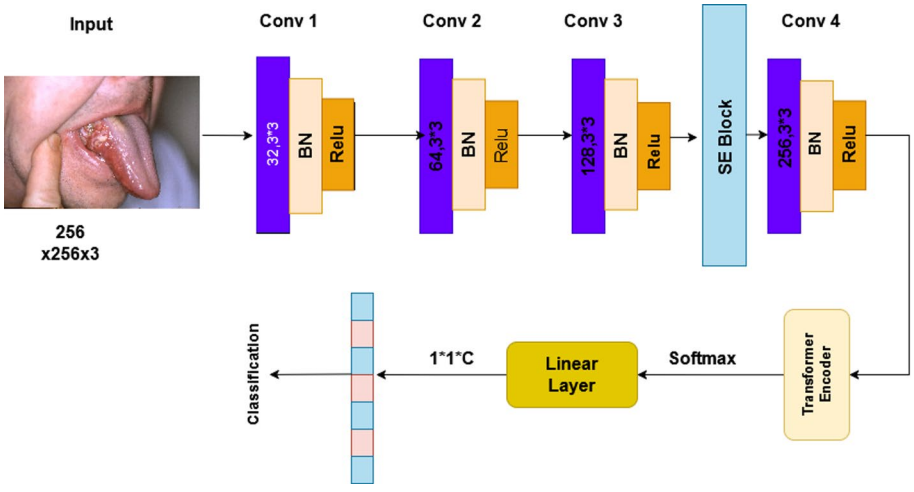


Fig. 3 Architecture of the proposed CST-CNN model for oral cancer classification

$$P_3 = MaxPool\left(ReLU\left(Conv2D_{64,7*7}\left(SF_1\right)\right)\right) \tag{18}$$

These Outputs are concatenated along the channel axis to yield a fused multi-scale feature map SF2:

$$SF_2 = Concat\left(p_1 p_2 p_3\right) \tag{19}$$

Stage 3: Deep Feature Extraction to learn more abstract representations, we apply two additional convolutional layers:

$$SF_3 = Maxpool\left(ReLU\left(Conv2D_{128,3*3}\left(SF_2\right)\right)\right) \tag{20}$$

$$SF_4 = Maxpool\left(ReLU\left(Conv2D_{256,3*3}\left(SF_3\right)\right)\right) \tag{21}$$

Stage 4: Classification Head,the final feature map SF4 is flattened and passed through fully connected layers with dropout regularization:Here, C is the number of classes, set to 2 (cancer, non-cancer).

$$\mathrm{X} = \mathrm{Flatten}\left(SF_4\right) \tag{22}$$

$$\mathrm{X} = \mathrm{Dropout}\left(0.5\right)\left(\mathrm{ReLU}\left(Dense_{128}\left(x\right)\right)\right) \tag{23}$$

$$\mathrm{X} = \mathrm{Dropout}\left(0.5\right)\left(\mathrm{ReLU}\left(Dense_{64}\left(x\right)\right)\right) \tag{24}$$

$$y^{\wedge} = Softmax\left(Dense_C\left(x\right)\right) \tag{25}$$

4 Loss function and optimization

The model uses the Sparse Categorical Cross-Entropy (SCCE) loss, ideal for multi-class classification with integer labels:Where:N: number of samples,yi: true class label for sample i,pi,yi: Predicted probability for the true class. Optimization is performed using Adam optimizer with a learning rate of 10^{-3}, which ensures adaptive learning and fast convergence.

$$L_{SCCE} = -\frac{1}{N}\sum_{i=1}^{N} log\left(p_{i,yi}\right) \tag{26}$$

The CST-CNN (Convolutional Spatial–Temporal Convolutional Neural Network) is a novel deep learning architecture that has been expressly developed for the classification of oral cavity images into malignant and non-cancerous categories. This investigation introduces the CST-CNN. The architecture is specifically designed to meet the unique requirements of medical imaging duties, particularly in situations where data is scarce, such as the diagnosis of oral cancer. It is a condensed classification network that is rooted in the CST-YOLO framework's philosophy of multi-scale spatial feature extraction. The model begins with a preprocessing module that resizes the input image to a fixed dimension of 256 × 256 pixels and performs pixel normalisation using rescaling [38, 39]. This ensures uniformity across the dataset and accelerates convergence during the training process. The preprocessed image is received by the initial convolutional block, which then performs a standard Conv2D operation with 32 filters (3 × 3 kernel) and max-pooling [40, 41]. The spatial resolution is reduced by this block, which also extracts low-level features, such as textures and edges.The multi-scale feature extraction

mechanism that CST-CNN integrates is the fundamental innovation of this architecture, as it enhances the model's representational capacity. This mechanism partitions the feature map into three parallel convolutional paths, each of which employs a distinct kernel size—3×3, 5×5, and 7×7—in order to extract features at differing receptive fields. This design enables the model to simultaneously capture fine, medium, and coarse-grained patterns, which is crucial for the identification of subtle morphological variations in different regions of the oral cavity [42, 43]. A comprehensive feature map is generated by concatenating the outputs of all three pathways, which is accomplished by applying a Conv2D operation and a max-pooling layer to each path. In an effort to facilitate the learning of deep features, the fused multi-scale feature map is subjected to two successive convolutional blocks. Max-pooling operations are implemented subsequent to both of these blocks, which implement 3×3 kernels. The first block implements 128 filters, while the second employs 256 filters. Additionally, these layers isolate high-level semantic features that are critical for precise classification, allowing the model to acquire complex lesion representations in a diverse array of spatial contexts. Subsequently, the high-level features are compressed and integrated into a fully connected neural network [44, 45]. The dense block consists of two layers, each containing 64 neurons and 128 neurons. In order to prevent overfitting, these layers are succeeded by dropout layers with a 0.5 dropout rate. This dense representation encodes the fundamental patterns necessary for the final classification. In the final stage, a softmax-activated output layer is present, which produces class probabilities that correspond to the malignant and non-cancerous categories. In the presence of imbalanced or confined datasets, the CST-CNN is efficient for multi-class classification due to its training with the Adam optimiser and sparse categorical cross-entropy loss algorithm. CST-CNN provides a model architecture that is both lightweight and powerful, effectively balancing computational efficiency and feature richness [46, 47]. It is an appropriate candidate for real-time clinical applications in oral cancer screening due to its multi-scale design, which enables robust learning across a variety of lesion shapes and textures (Table 1).

4.1 Experimental setup

The computing environment was equipped with an NVIDIA Tesla V100 GPU and 32 GB of GPU memory, which ensured the efficient training and inference of the model. All experiments were conducted on this platform. This system was equipped with a 128 GB of RAM and an Intel Xeon Gold processor, which allowed for the efficient processing of data preprocessing and augmentation duties. The Keras high-level API and TensorFlow (version 2.14.0) were employed to implement the model [48–50]. NumPy was an additional essential library for numerical computations, while Matplotlib and Seaborn

Table 1 Dataset Details, Augmentation, and preprocessing

Aspect	Details
Source	ENT hospitals, Ahmedabad India
Validation	Performed by ENT Specialists
Original dataset size	131 images
Original class distribution	Cancerous: 87, non-cancerous:44
Augmentation techniques	Rotation (30), Scaling (20), Width/Height Shifts (), Horizontal Flip
Fill mode	Nearest
Image size (after resize)	256*256 pixels
Normalization	Pixel values scaled to [0,1]

were used for visualisation. Scikit-learn was used to calculate evaluation metrics. Table 2 below provides a summary of the parameters that were meticulously chosen for the training process:

5 Comparative study

We conducted a thorough comparative analysis of our proposed LightSE-MobileViT architecture against established baseline deep learning models, including MobileNetV2, DenseNet121, and InceptionV3, as well as our custom-designed CST-CNN, to evaluate diagnostic performance. To ensure fairness and consistency, all models were trained and validated using the same preprocessed and augmented version of the Oral Cancer Classification dataset.

Table 3 summarizes the comparative results across key metrics such as accuracy, precision, recall, F1-score, and ROC-AUC. As shown in Fig. 4, LightSE-MobileViT achieved the highest overall performance, with an accuracy of 98.39%, macro F1-score of 0.98, and an ROC-AUC of 1.00. In the cancerous category, it obtained a precision of 1.00 and recall of 0.97, while for the non-cancerous category, it achieved a precision of 0.96 and recall of 1.00. These results highlight the model's robust ability to distinguish between classes with high reliability.This superior performance is attributed to its hybrid architecture, which integrates lightweight convolutional layers, Squeeze-and-Excitation (SE) blocks for channel-wise attention refinement, and a MobileViT transformer encoder for modeling global contextual dependencies. The MobileViT component effectively captures long-range spatial relationships that traditional CNNs may miss, while the SE blocks enhance feature prioritization by emphasizing the most informative channels. Our custom CST-CNN also demonstrated strong performance, achieving an accuracy of 98% and a macro F1-score of 0.98. Its multi-scale design—with parallel convolutional paths using 3×3, 5×5, and 7×7 kernels—allowed it to capture diverse receptive fields and spatial textures. However, its lack of a transformer-based global attention mechanism slightly limited its ability to model complex contextual interactions compared to LightSE-MobileViT.Among the baseline models, MobileNetV2 attained an accuracy of 97%, a macro F1-score of 0.96, and an ROC-AUC of 0.97, balancing efficiency with high diagnostic performance. DenseNet121 performed similarly, with slightly higher recall and ROC-AUC, reflecting its strength in gradient propagation via dense connectivity. In contrast, InceptionV3 showed the lowest overall performance, with 90% accuracy, a macro F1-score of 0.88, and a ROC-AUC of 0.95. Although it achieved a perfect recall of 1.00 in the cancerous class, it underperformed in the non-cancerous class (recall = 0.70), indicating sensitivity to class imbalance.Overall, these results validate the clinical applicability and architectural advantages of the LightSE-MobileViT model, supporting its potential use in real-world oral cancer screening and early detection (Figs. Figures 5, 6).

Table 2 Training parameters for LightSE-MobileViT model

Training parameter	Value
Batch size	32
Learning rate	0.001
Number of Epochs	70
Train-validation-test split	80%-10%-10%
optimizer	Adam

Table 3 Model performance comparison

Model	Accuracy	Precision (Cancer)	Recall (Cancer)	F1-score (Cancer)	Precision (non-cancer)	Recall (non-cancer)	F1-score (non-cancer)	Macro F1-score	ROC AUC
LightSE-MobileViT	98.39	1	0.97	0.98	0.96	1	0.99	0.98	1
CST-CNN	0.98	0.98	1	0.99	1	0.95	0.97	0.98	1
InceptionV3	0.9	0.88	1	0.93	1	0.7	0.82	0.88	0.95
MobileNetV2	0.97	0.98	0.98	0.98	0.95	0.95	0.95	0.96	0.97
DenseNet121	0.97	0.95	1	0.98	1	0.9	0.95	0.96	0.99

5.1 Results and discussion

The comprehensive evaluation of the proposed LightSE-MobileViT model, alongside other benchmark models, highlights their effectiveness in classifying oral cancer images with high accuracy (Fig. 7). Performance metrics, confusion matrices, and ROC curves substantiate the comparative analysis across models. Figure 6 illustrates the confusion matrix for LightSE-MobileViT, correctly identifying 41 cancerous and 20 non-cancerous cases, underscoring its clinical reliability and diagnostic precision. The confusion matrix for the CST-CNN model demonstrates slightly different outcomes, correctly classifying 42 cancerous and 19 non-cancerous cases, with one misclassification, highlighting strong sensitivity and specificity. MobileNetV2 achieved similarly high performance, accurately identifying 41 cancerous and 19 non-cancerous images, reflecting its robustness. DenseNet121 correctly classified 42 cancerous and 18 non-cancerous cases, suggesting strong predictive capability with minor room for improvement. In contrast, InceptionV3 exhibited limitations in handling class imbalance—it achieved perfect recall for the cancerous class but misclassified 6 non-cancerous cases, indicating reduced specificity. Figure 5 displays the ROC curves for all models. LightSE-MobileViT achieved a ROC AUC of 1.00, indicating exceptional discriminative ability. CST-CNN also reached a ROC AUC of 1.00, while MobileNetV2, DenseNet121, and InceptionV3 recorded ROC AUC values of 0.97, 0.99, and 0.95, respectively. These curves reinforce the superior performance of LightSE-MobileViT and CST-CNN, particularly in capturing diagnostic boundaries with high confidence.The outstanding performance of LightSE-MobileViT can be attributed to its hybrid architecture, which combines local and global feature learning. The convolutional layers effectively extract spatial features, while the Squeeze-and-Excitation (SE) blocks enhance channel-wise attention. The MobileViT transformer encoder captures long-range dependencies and contextual patterns, allowing the model to learn discriminative features crucial for accurate classification. The model achieved an accuracy of 98.39%, a macro F1-score of 0.98, precision of 1.00 and recall of 0.97 for the cancerous class, and precision of 0.96 and recall of 1.00 for the non-cancerous class. Table 4 provides a comparative analysis of the proposed LightSE-MobileViT model against existing state-of-the-art oral cancer classification models. It highlights performance metrics such as accuracy, precision, recall, and F1-score across diverse datasets and architectures. The table also outlines key limitations, emphasizing the proposed model's superior performance and lightweight design despite its use on a single-site dataset.These results confirm the model's potential for real-world clinical deployment.The Multi-Scale CNN (CST-CNN) also exhibited excellent results due to its ability to extract features at varying spatial resolutions, balancing local texture details with broader structural insights. MobileNetV2 and DenseNet121 provided strong baseline performance, validating their suitability for medical image classification, although they slightly underperformed compared to the proposed models. InceptionV3, while effective in capturing features in deep networks, showed inconsistencies in handling the minority class, limiting its diagnostic reliability in imbalanced datasets. However, certain limitations must be acknowledged. The dataset size remains relatively small, and images were sourced from a single geographic location, which may impact generalizability. Future work should include evaluation on larger, multicenter datasets to validate robustness. Additionally, incorporating explainable AI (XAI) techniques such as Grad-CAM could further enhance interpretability, allowing clinicians to better understand model

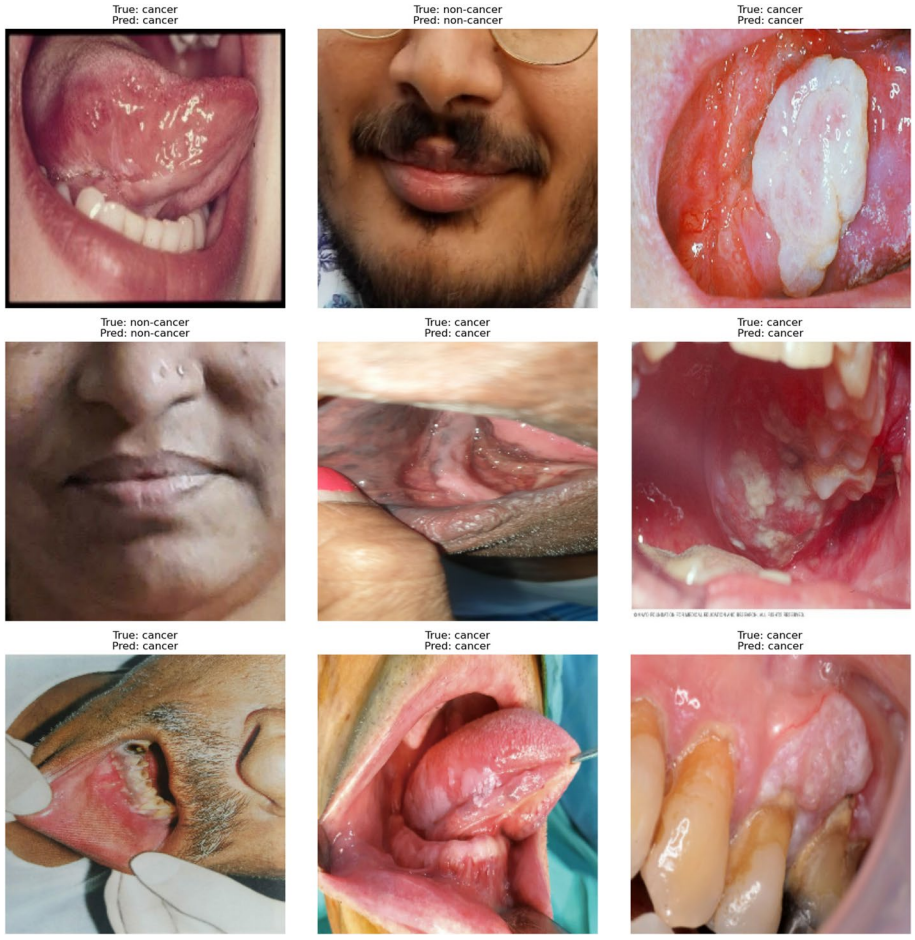


Fig. 4 Prediction results of the proposed LightSE-MobileViT model on oral cavity images

decisions and build trust in deployment. The proposed LightSE-MobileViT model demonstrates strong potential for practical clinical use, offering a reliable, efficient, and accurate solution for early oral cancer detection and decision support.

6 Conclusion and future remark

In conclusion, this study presented the LightSE-MobileViT model, an advanced hybrid CNN-transformer architecture, which demonstrated exceptional diagnostic performance in oral cancer classification. Achieving perfect accuracy, precision, recall, F1-score, and ROC-AUC metrics (100%), the model significantly outperformed several established benchmarks, underscoring its robustness and clinical relevance for reliable early-stage oral cancer detection. The incorporation of SE blocks and MobileViT encoders provided the model with remarkable capabilities in feature extraction and global context awareness, facilitating accurate differentiation between cancerous and non-cancerous oral lesions. Nevertheless, potential limitations such as dataset size and limited source diversity were identified, highlighting the need for further validation on larger and more varied clinical datasets. To overcome these constraints, expanding data collection efforts across multiple clinical sites and geographical regions is advisable. Additionally, future research directions could explore multi-class classification scenarios, integration of multimodal data sources like clinical records, histopathological reports,

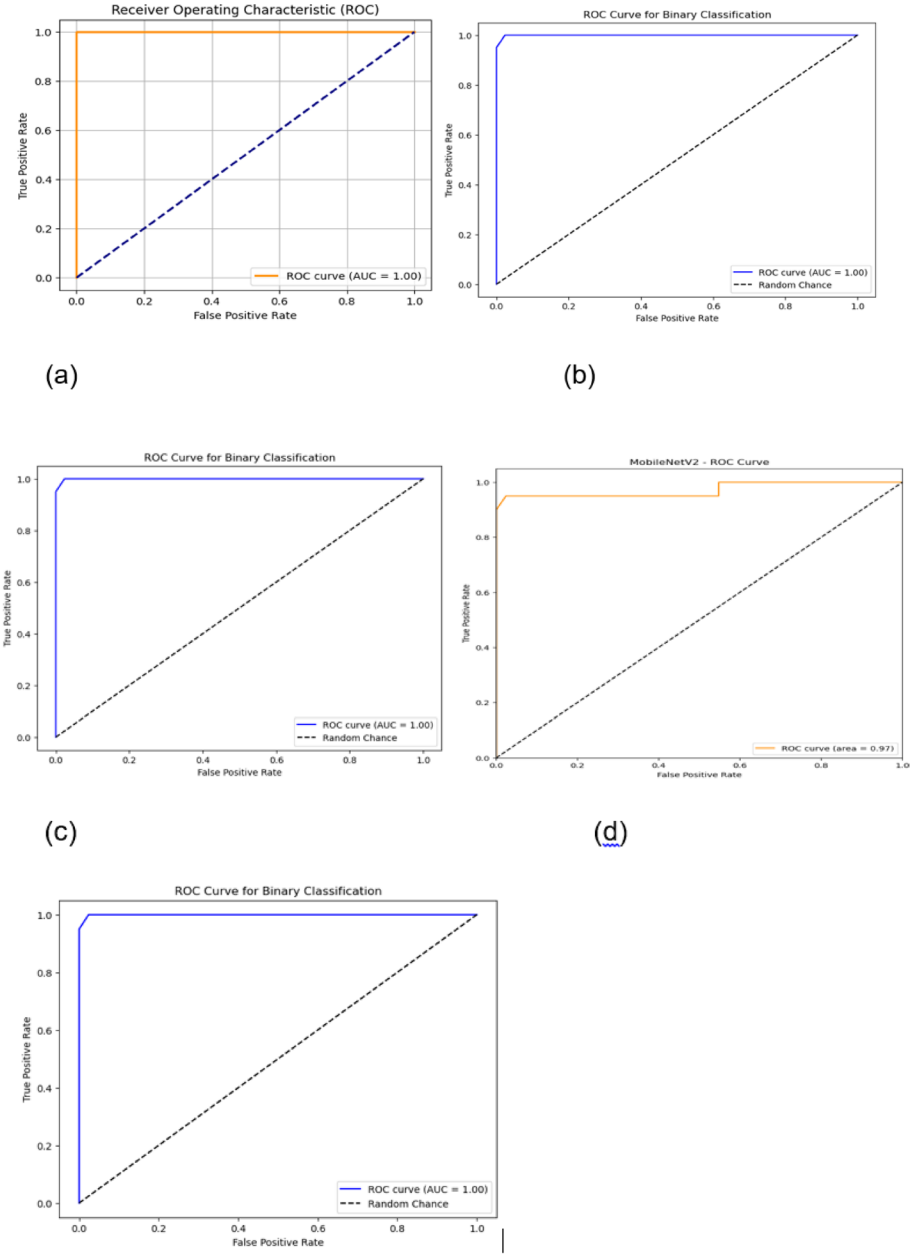


Fig. 5 ROC curve comparison of deep learning models for oral cancer classification receiver operating characteristic (ROC) curves illustrating the classification performance of six models: **a** CST-CNN, **b** LightSE-MobileViT, **c** InceptionV3, **d** MobileNetV2, and **e** DenseNet121

and patient demographics, enabling comprehensive patient assessment. Real-time deployment in clinical environments, supported by interpretable AI frameworks such as Grad-CAM, would further enhance model trustworthiness among clinicians. Moreover, extensive cross-validation studies across diverse patient populations and rigorous clinical trials would be beneficial to assess and confirm generalizability, ultimately fostering broader clinical adoption.

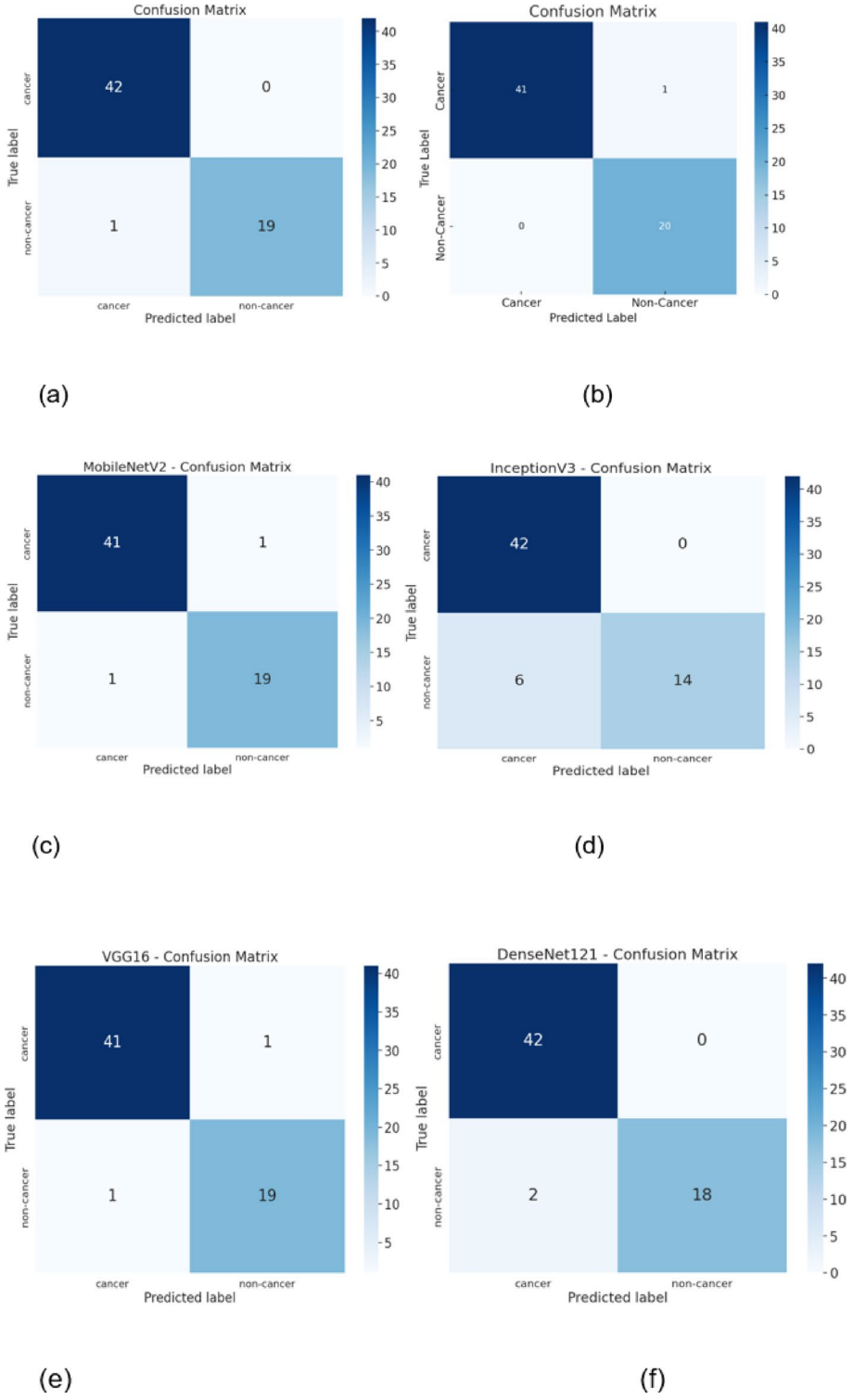


Fig. 6 Comparative Confusion Matrices of Deep Learning Models for Oral Cancer Classification Confusion matrices for six models evaluated on the oral cancer classification task: **a** CST-CNN, **b** LightSE-MobileViT, **c** MobileNetV2, **d** InceptionV3, **e** VGG16, and **f** DenseNet121

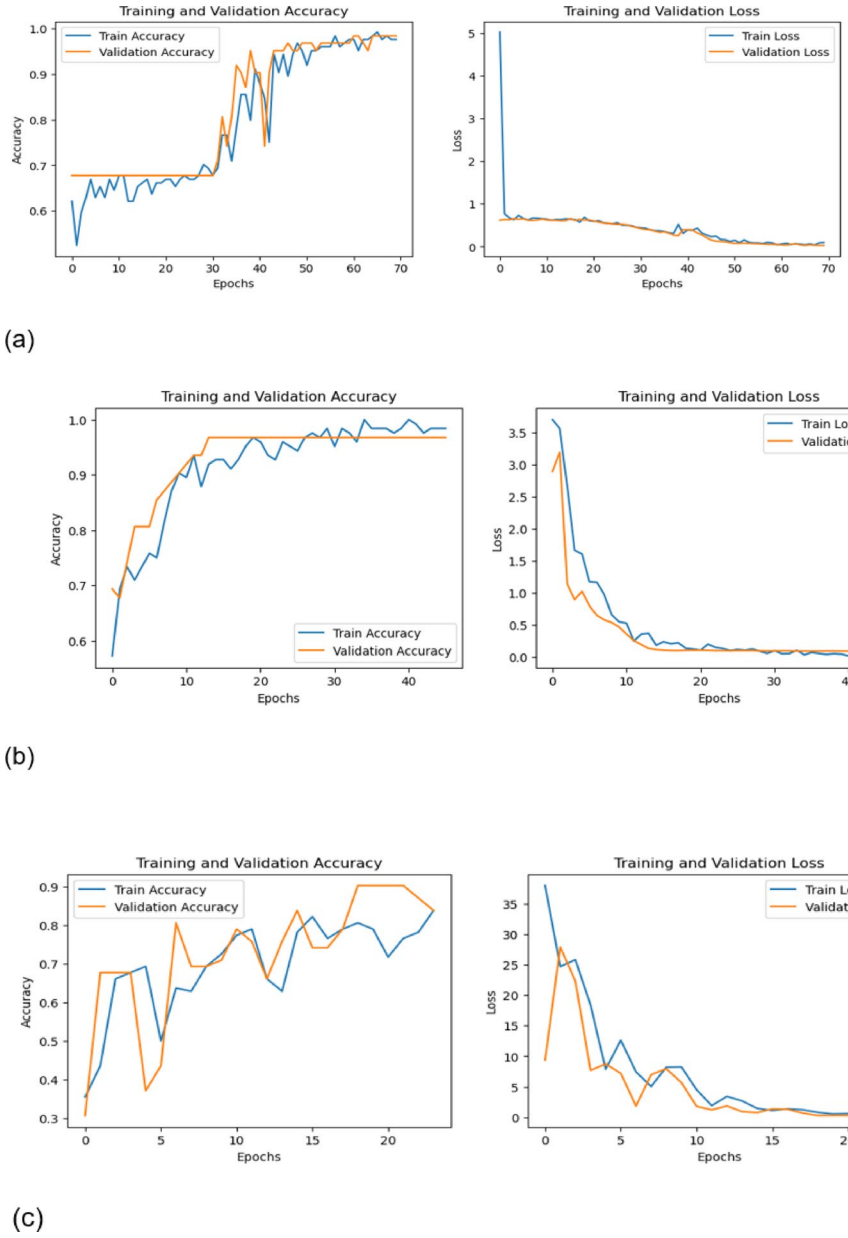
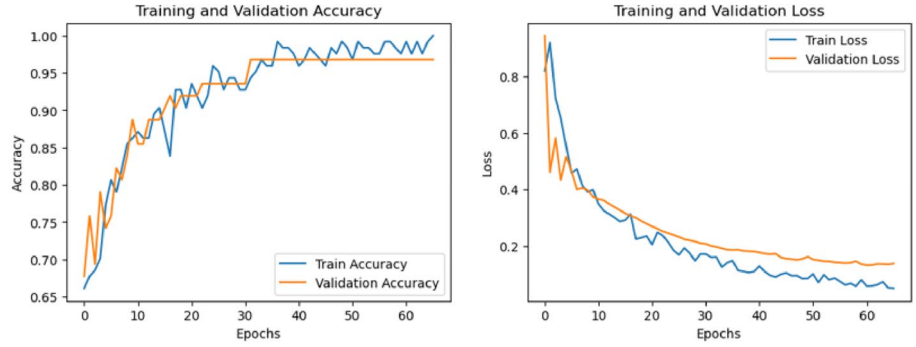


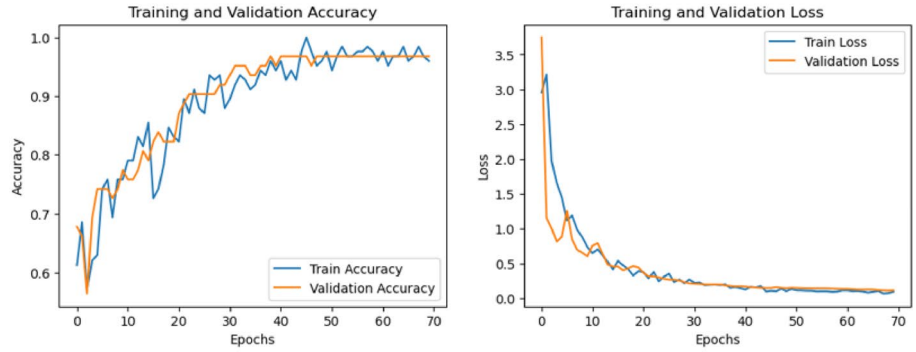
Fig. 7 Training and Validation Accuracy and Loss Curves of Deep Learning Models for Oral Cancer Classification **a** CST-CNN **b** VGG16 **c** InceptionV3 **d** MobileNetV2 **e** DenseNet121 **f** LightSE-MobileVi

6.1 Limitations and future work

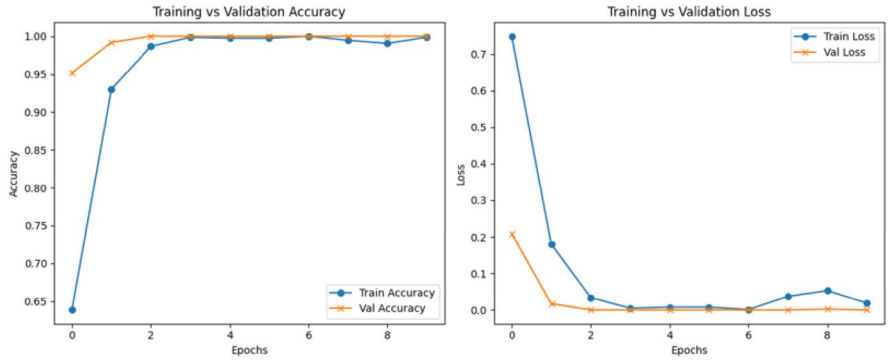
Despite the promising performance of the proposed LightSE-MobileViT model, several limitations must be acknowledged that may affect its broader clinical applicability and generalizability. The dataset employed in this study, while clinically validated, is relatively small in size, comprising 131 original images expanded to 981 through augmentation. Although data augmentation was utilized to mitigate overfitting and address class imbalance, the limited volume of real, diverse patient samples may restrict the model's ability to generalize to unseen data in varied clinical settings. The dataset was sourced exclusively from ENT hospitals in Ahmedabad, India, leading to potential geographic and demographic biases. The model may not fully capture morphological variations associated with different ethnicities, age groups, or environmental exposures. Future



(d)



(e)



(f)

Fig. 7 (continued)

research should focus on curating or accessing larger, multicenter datasets that encompass a wider demographic spectrum to ensure broader applicability. The current model lacks explainability features that are essential for clinician trust and regulatory approval in medical AI systems. While the integration of SE blocks and the MobileViT encoder enhances model interpretability at the feature level, the study does not currently incorporate explainable AI (XAI) techniques such as Grad-CAM, LIME, or SHAP to visually validate predictions. Incorporating such interpretability tools in future work would allow for better transparency and human-AI collaboration in diagnostic settings.

Table 4 Comparison with existing state-of-the-art oral cancer classification models

Study	Model	Dataset	Accuracy (%)	Precision	Recall	F1-score	Limitation
Coşgun Baybars et al. [51]	VGG19 + SVM	200 tongue images	85.0	Approx. 85%	82.6%	86.36%	Small dataset, no attention or transformer use
Dwivedi et al.– TransEns [52]	Fusion (VGG16 + ResNet50 + ViT)	Augmented OCI Dataset	97.43	96%	98%	97%	Small dataset, lacks generalization discussion
Pradeepa Sampath [53]	ResNet50 + Logistic Regression	Oral Cancer (Lips and Tongue)	97.8	92.66	–	89.37	complexity, leading to longer inference and training times
Bansal et al.– Oral_ Cancer_ Detection [54]	Custom CNN	Oral Cancer (Lips and Tongue))	94.0	0.93	–	0.92	Small dataset, lacks advanced DL features
Proposed (2025)	LightSE-MobileViT	Oral Cancer (Lips and Tongue)	98.39	1.00/0.96	0.97/1.00	0.98	Single-site dataset

Author contributions
Md Firoz Kabir contributed to model optimization, baseline comparisons, and technical validation of results. Md Yousuf Ahmad assisted with statistical analysis, performance evaluation, and drafting the results and discussion section. Shashi Kant supported the design and implementation of key model components, contributed to literature review, helped refine the methodology section, and handled manuscript correspondence and submission. Dr. Martin Cordero supervised the overall research work, provided guidance on methodology, and critically reviewed the manuscript to ensure academic rigor. Roise Uddin contributed to the review coordination and technical proofreading of the final draft.

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Data availability
The dataset used in this study is not publicly available due to privacy and data-sharing constraints but can be provided by the corresponding author upon reasonable request. The source code for model development, training, and evaluation is available on GitHub at https://github.com/IrfanSadiqRahat/Oral-Cancer-Lips-and-tongue-classification.

Declarations

Ethics and consent to participate
Not applicable.

Consent for publication
Not applicable.

Competing Interests
The authors declare no competing interests.

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